KERALA POLYTECHNIC · REVISION 2026

Electric and Hybrid Vehicles

Complete student lesson for course code **6051A**.

Revision 2026 Semester 6 Program Elective 4 modules

Course snapshot

Scheme: 4 (L:4, T:0, P:0)

Credits: 4

Contact: 60 periods per semester

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: The supplied PDF does not specify a separate CIE/ESE distribution. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

6051A

Semester

6

Credits

4

Teaching scheme

4 (L:4, T:0, P:0)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Electric and Hybrid Vehicle Fundamentals	12	CO1
2	EV Design and Motor Drives	16	CO2
3	Power Electronics, Drive Trains and Braking	14	CO3
4	Energy Storage, Fuel Cells and Charging	16	CO4

Prerequisites

- Knowledge on basic Automobile Engineering — Basic Automobile Engineering, Semester 2.
- Knowledge on Automobile electrical and electronics systems — Automobile Electrical and Electronic systems, Semester 4.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. To understand general aspects of Electric and Hybrid Vehicles.
2. To understand basic design considerations of EVs.
3. To select the suitable electric propulsion systems.
4. To select required energy storage and charging devices.

Course Outcomes

1. Summarize the general aspects of Electric and Hybrid Vehicles.
2. Outline the basic design considerations of Electric vehicles.
3. Identify various drive trains in Electric vehicles and Hybrid electric vehicles.
4. Select the required energy storage and charging devices for Electric and Hybrid vehicles.

CO–PO mapping as supplied

The supplied CO–PO table is reproduced at outcome level from the PDF; blank cells are not treated as mapped.

Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Need for hybrid and electric vehicles	1
module-1	M1.02	Components and working principle of electric and hybrid electric vehicles	3
module-1	M1.03	Configurations of electric vehicles and hybrid vehicles	4
module-1	M1.04	Compare petrol, diesel, hybrid and electric vehicles	2
module-1	M1.05	Advantages and disadvantages of hybrid and electric vehicles	1
module-1	M1.06	Government policies and schemes related to Electric vehicles	1
module-2	M2.01	Design requirements for electric vehicles	3
module-2	M2.02	Performance during acceleration, coasting, moving up and down a hill	5
module-2	M2.03	Types of motors used in electric vehicles	3
module-2	M2.04	Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives	5
module-3	M3.01	Power electronics converters used in electric and hybrid vehicles	2
module-3	M3.02	Control systems for EV	2
module-3	M3.03	Drive trains in Electric and Hybrid Electric Vehicles	2
module-3	M3.04	Drive train configurations in electric and hybrid electric vehicles	3

Module	Topic code	Topic	Hours
module-3	M3.05	Regenerative braking and hybrid braking	2
module-3	M3.06	Service and maintenance of EV components	3
module-4	M4.01	Energy storage requirements in electric and hybrid vehicles	2
module-4	M4.02	Construction and working of batteries used in electric and hybrid vehicles	5
module-4	M4.03	Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages	3
module-4	M4.04	Operation of Fuel cells	2
module-4	M4.05	Battery charging Technologies and Battery management systems	4

Learning Roadmap

1. Electric and Hybrid Vehicle Fundamentals
CO1 · 12 hours

2. EV Design and Motor Drives
CO2 · 16 hours

3. Power Electronics, Drive Trains and Braking
CO3 · 14 hours

4. Energy Storage, Fuel Cells and Charging
CO4 · 16 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Electric and Hybrid Vehicle Fundamentals

This module introduces why electric and hybrid propulsion is required, the main vehicle configurations, comparative performance, and the policy context.

Hours: 12

CO1 · Bloom level: Understanding

Learning goals

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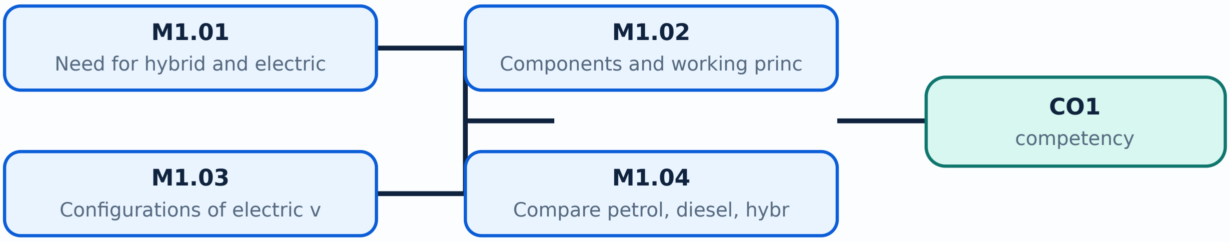
Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

M1.01 — Need for hybrid and electric vehicles (1 hours)

Concept and explanation

<p>Scope. The official syllabus places Need for hybrid and electric vehicles in this topic. The required scope is: Need for hybrid and electric vehicles; historical background; social and environmental importance.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Need for hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Need for hybrid and electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Need for hybrid and electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Need for hybrid and electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Need for hybrid and electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Components and working principle of electric and hybrid electric vehicles in this topic. The required scope is: Main components and working principles of electric and hybrid vehicles.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Components and working principle of electric and hybrid electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Components and working principle of electric and hybrid electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Components and working principle of electric and hybrid electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Components and working principle of electric and hybrid electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Components and working principle of electric and hybrid electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours)

Concept and explanation

<p>Scope. The official syllabus places Configurations of electric vehicles and hybrid vehicles in this topic. The required scope is: Pure Electric vehicle (PE), Battery Electric vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric vehicle (HEV), Plug-in Hybrid Electric vehicle (PHEV), and types based on differential; series, parallel, series-parallel and complex HEV.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Configurations of electric vehicles and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Configurations of electric vehicles and hybrid vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Configurations of electric vehicles and hybrid vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Configurations of electric vehicles and hybrid vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Configurations of electric vehicles and hybrid vehicles without explaining the working sequence, variables, or engineering reason for each item.

M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Compare petrol, diesel, hybrid and electric vehicles in this topic. The required scope is: Comparative study of EV and conventional vehicles; differences between complete EV and hybrid vehicles; life cycle assessment.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Compare petrol, diesel, hybrid and electric vehicles</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Compare petrol, diesel, hybrid and electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Compare petrol, diesel, hybrid and electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Compare petrol, diesel, hybrid and electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Compare petrol, diesel, hybrid and electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours)

Concept and explanation

<p>Scope. The official syllabus places Advantages and disadvantages of hybrid and electric vehicles in this topic. The required scope is: Technical, economic, environmental, range, charging, maintenance, and lifecycle advantages and limitations.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Advantages and disadvantages of hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Advantages and disadvantages of hybrid and electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Advantages and disadvantages of hybrid and electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Advantages and disadvantages of hybrid and electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Advantages and disadvantages of hybrid and electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

M1.06 — Government policies and schemes related to Electric vehicles (1 hours)

Concept and explanation

<p>Scope. The official syllabus places Government policies and schemes related to Electric vehicles in this topic. The required scope is: FAME-1, FAME-2, Transformative Mobility and Energy Storage Mission, central and state policies, subsidies, incentives, and global experiences.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Government policies and schemes related to Electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Government policies and schemes related to Electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.

- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Government policies and schemes related to Electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Government policies and schemes related to Electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Government policies and schemes related to Electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Module summary: Electric vehicle architecture and policy comparison.

OFFICIAL MODULE 2

EV Design and Motor Drives

Design begins with the required range, speed, acceleration, gradeability, mass, tractive effort, transmission efficiency, and energy consumption.

Hours: 16
CO2 · Bloom level: Understanding

Learning goals

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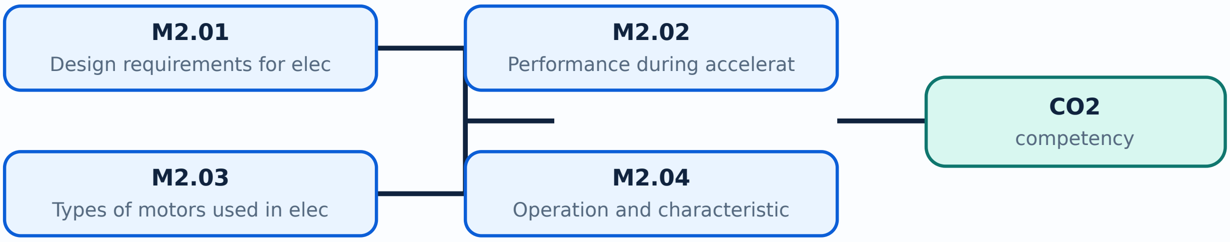
Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

M2.01 — Design requirements for electric vehicles (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Design requirements for electric vehicles in this topic. The required scope is: Range, maximum velocity, acceleration, power requirement, mass, resistances, tractive effort, transmission requirement, transmission efficiency and energy consumption.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Design requirements for electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr> <th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. </td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody> </table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Malayalam support: Malayalam support: Design requirements for electric vehicles □□□□ topic □□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Design requirements for electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Design requirements for electric vehicles is applied in electric propulsion systems practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Design requirements for electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Design requirements for electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours)

Concept and explanation

<p>Scope. The official syllabus places Performance during acceleration, coasting, moving up and down a hill in this topic. The required scope is: Performance of electric vehicles during acceleration, coasting, and motion up or down a hill. </p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Performance during acceleration, coasting, moving up and down a hill</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Performance during acceleration, coasting, moving up and down a hill □□□□ topic □□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Performance during acceleration, coasting, moving up and down a hill using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Performance during acceleration, coasting, moving up and down a hill is applied in electric propulsion systems practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Performance during acceleration, coasting, moving up and down a hill**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Performance during acceleration, coasting, moving up and down a hill without explaining the working sequence, variables, or engineering reason for each item.

M2.03 — Types of motors used in electric vehicles (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Types of motors used in electric vehicles in this topic. The required scope is: Classification of EV motors and selection considerations.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Types of motors used in electric vehicles</caption><thead><tr> <th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th> <td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Types of motors used in electric vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Types of motors used in electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Types of motors used in electric vehicles is applied in electric propulsion systems practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Types of motors used in electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Types of motors used in electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours)

Concept and explanation

<p>Scope. The official syllabus places Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives in this topic. The required scope is: Configuration and control of DC motor drives, induction motor drives, permanent magnet motor drives, and switched reluctance motor drives.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr> <th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. </td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody> </table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Malayalam support: Malayalam support: Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.

- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives is applied in electric propulsion systems practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives without explaining the working sequence, variables, or engineering reason for each item.

Module summary: Use tractive effort and motor characteristics to justify a drive choice.

OFFICIAL MODULE 3

Power Electronics, Drive Trains and Braking

Power converters, control systems, drive-train architecture, regenerative braking, and service practice connect the energy source to the road wheels.

Hours: 14
CO3 · Bloom level: Understanding

Learning goals

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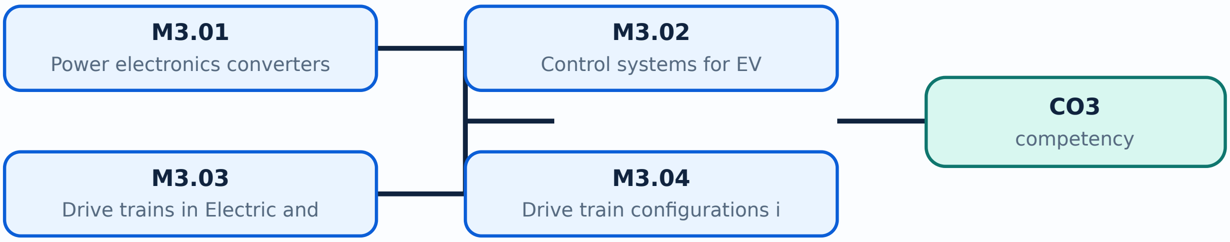
Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Power electronics converters used in electric and hybrid vehicles in this topic. The required scope is: Rectifiers, inverters, DC/DC converters, and power-split devices for hybrid vehicles.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Power electronics converters used in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Power electronics converters used in electric and hybrid vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Power electronics converters used in electric and hybrid vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Power electronics converters used in electric and hybrid vehicles is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Power electronics converters used in electric and hybrid vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Power electronics converters used in electric and hybrid vehicles without explaining the working sequence, variables, or engineering reason for each item.

M3.02 — Control systems for EV (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Control systems for EV in this topic. The required scope is: Sizing the drive system, propulsion motor, power electronics, energy-storage technology, communications, and supporting subsystems. </p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Control systems for EV</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. </td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody> </table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Control systems for EV □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Control systems for EV using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Control systems for EV is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Control systems for EV**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Control systems for EV without explaining the working sequence, variables, or engineering reason for each item.

M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Drive trains in Electric and Hybrid Electric Vehicles in this topic. The required scope is: Basic architecture, general EV configuration, alternatives based on drive-train and power-source configuration, single/multi-motor and in-wheel drives.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Drive trains in Electric and Hybrid Electric Vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Malayalam support: Malayalam support: Drive trains in Electric and Hybrid Electric Vehicles □□□□ topic □□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Drive trains in Electric and Hybrid Electric Vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Drive trains in Electric and Hybrid Electric Vehicles is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Drive trains in Electric and Hybrid Electric Vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Drive trains in Electric and Hybrid Electric Vehicles without explaining the working sequence, variables, or engineering reason for each item.

M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Drive train configurations in electric and hybrid electric vehicles in this topic. The required scope is: Series, parallel, series-parallel, and complex hybrid drive trains.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Drive train configurations in electric and hybrid electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Drive train configurations in electric and hybrid electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Drive train configurations in electric and hybrid electric vehicles is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Drive train configurations in electric and hybrid electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Drive train configurations in electric and hybrid electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

M3.05 — Regenerative braking and hybrid braking (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Regenerative braking and hybrid braking in this topic. The required scope is: Braking in electric and hybrid vehicles, energy recovery and coordination with friction braking.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Regenerative braking and hybrid braking</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Malayalam support: Malayalam support: Regenerative braking and hybrid braking □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Regenerative braking and hybrid braking using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Regenerative braking and hybrid braking is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Regenerative braking and hybrid braking**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Regenerative braking and hybrid braking without explaining the working sequence, variables, or engineering reason for each item.

M3.06 — Service and maintenance of EV components (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Service and maintenance of EV components in this topic. The required scope is: Maintenance, repair, service, and troubleshooting of motor, drive train, battery, and other EV components.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Service and maintenance of EV components</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Service and maintenance of EV components ▯▯▯▯ topic ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ purpose, working principle, ▯▯▯▯▯▯ components ▯▯▯▯▯▯▯▯▯▯ variables, ▯▯▯▯▯▯ performance ▯▯▯▯▯▯▯▯▯▯ safety checks ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯. Technical terms English-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯; diagram ▯▯▯▯▯▯▯▯▯▯ step sequence ▯▯▯▯▯▯▯▯▯▯ process ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯.

Study points

- Define Service and maintenance of EV components using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Service and maintenance of EV components is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Service and maintenance of EV components**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Service and maintenance of EV components without explaining the working sequence, variables, or engineering reason for each item.

Module summary: Trace power flow from battery through converter and motor to wheels, including regenerative return.

OFFICIAL MODULE 4

Energy Storage, Fuel Cells and Charging

Energy storage determines range, power capability, charging time, safety, cost, and lifecycle of an EV or HEV.

Hours: 16
CO4 · Bloom level: Understanding

Learning goals

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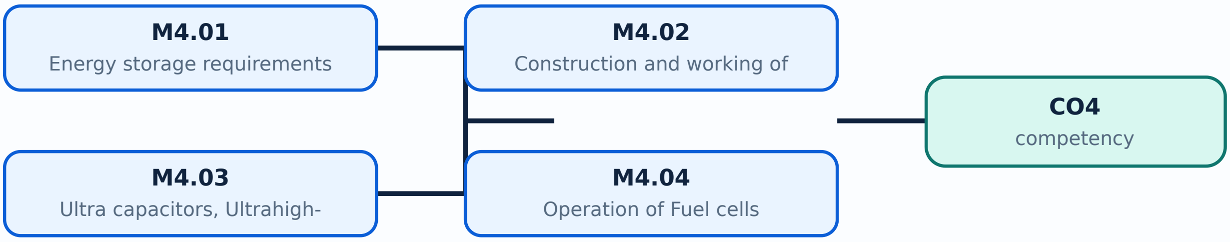
Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Energy storage requirements in electric and hybrid vehicles in this topic. The required scope is: Energy-storage requirements and battery parameters for EV and HEV applications.</p> <p> Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Energy storage requirements in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th> <th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr> <tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody> </table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Energy storage requirements in electric and hybrid vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Energy storage requirements in electric and hybrid vehicles is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Energy storage requirements in electric and hybrid vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Energy storage requirements in electric and hybrid vehicles without explaining the working sequence, variables, or engineering reason for each item.

M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours)

Concept and explanation

<p>Scope. The official syllabus places Construction and working of batteries used in electric and hybrid vehicles in this topic. The required scope is: Lead-acid, nickel-based, sodium-based, lithium-based, and metal-air batteries; construction, operation, and characteristics.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Construction and working of batteries used in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr> <tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Construction and working of batteries used in electric and hybrid vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Construction and working of batteries used in electric and hybrid vehicles is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Construction and working of batteries used in electric and hybrid vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Construction and working of batteries used in electric and hybrid vehicles without explaining the working sequence, variables, or engineering reason for each item.

M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages in this topic. The required scope is: Features and principle of ultracapacitors, flywheel storage, and hybridization of storage devices.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr> <tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

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Study points

- Define Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages without explaining the working sequence, variables, or engineering reason for each item.

M4.04 — Operation of Fuel cells (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Operation of Fuel cells in this topic. The required scope is: Basic principle, operation, and types of fuel cells.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Operation of Fuel cells</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr> <tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p> Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Malayalam support: Malayalam support: Operation of Fuel cells □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Operation of Fuel cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Operation of Fuel cells is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Operation of Fuel cells**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Operation of Fuel cells without explaining the working sequence, variables, or engineering reason for each item.

M4.05 — Battery charging Technologies and Battery management systems (4 hours)

Concept and explanation

<p>Scope. The official syllabus places Battery charging Technologies and Battery management systems in this topic. The required scope is: Onboard/offboard charging, domestic/public infrastructure, fast charging, battery swapping, and battery-management systems.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"> <table><caption>Study frame for Battery charging Technologies and Battery management systems</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr> <tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Battery charging Technologies and Battery management systems □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Battery charging Technologies and Battery management systems using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Battery charging Technologies and Battery management systems is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Battery charging Technologies and Battery management systems**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Battery charging Technologies and Battery management systems without explaining the working sequence, variables, or engineering reason for each item.

Module summary: Select storage and charging technology against range, power, safety, and infrastructure constraints.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

ule to view its labelled learning-map diagram.

Module 2: EV Design and

rning-map diagram.

Module 3: Power

its labelled learning-map diagram.

Module 4: Energy Storage,

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list for this theory course. Use the topic procedures, worked examples, diagrams, and module questions in this lesson for structured study.

Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment for this theory course.

Assessment Methodology

The supplied PDF does not specify a separate CIE/ESE distribution.

- The supplied PDF lists Series Test – I and Series Test – II entries, but a complete official CIE/ESE mark distribution and question-paper pattern are not specified in the supplied PDF.
- The practice model paper in this lesson is therefore explicitly a study aid, not an official examination paper.

Module-wise Questions and Answers

module-1 — Electric and Hybrid Vehicle Fundamentals

1. Explain Need for hybrid and electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Need for hybrid and electric vehicles in this topic. The required scope is: Need for hybrid and electric vehicles; historical background; social and environmental importance.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the

transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition. </p> <div class="table-wrap"><table><caption>Study frame for Need for hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr> </thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th> <td>How the input is converted, controlled, transported, stored, measured, or protected. </td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr> <th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p> Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Components and working principle of electric and hybrid electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Components and working principle of electric and hybrid electric vehicles in this topic. The required scope is: Main components and working principles of electric and hybrid vehicles.</p> <p> Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition. </p> <div class="table-wrap"><table><caption>Study frame for Components and working principle of electric and hybrid electric vehicles</caption><thead><tr><th>Aspect</th> <th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering. </td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th> <td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td> </tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes

normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Configurations of electric vehicles and hybrid vehicles. **[3 marks]**

<p>Scope. The official syllabus places Configurations of electric vehicles and hybrid vehicles in this topic. The required scope is: Pure Electric vehicle (PE), Battery Electric vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric vehicle (HEV), Plug-in Hybrid Electric vehicle (PHEV), and types based on differential; series, parallel, series-parallel and complex HEV.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Configurations of electric vehicles and hybrid vehicles</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Compare petrol, diesel, hybrid and electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Compare petrol, diesel, hybrid and electric vehicles in this topic. The required scope is: Comparative study of EV and conventional vehicles; differences between complete EV and hybrid vehicles; life cycle assessment.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Compare petrol, diesel, hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — EV Design and Motor Drives

5. Explain Design requirements for electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Design requirements for electric vehicles in this topic. The required scope is: Range, maximum velocity, acceleration, power requirement, mass, resistances, tractive effort, transmission requirement, transmission efficiency and energy consumption.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Design requirements for electric

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Performance during acceleration, coasting, moving up and down a hill. **[3 marks]**

<p>Scope. The official syllabus places Performance during acceleration, coasting, moving up and down a hill in this topic. The required scope is: Performance of electric vehicles during acceleration, coasting, and motion up or down a hill. </p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Performance during acceleration, coasting, moving up and down a hill</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Types of motors used in electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Types of motors used in electric vehicles in this topic. The required scope is: Classification of EV motors and selection considerations.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Types of motors used in electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives. **[3 marks]**

<p>Scope. The official syllabus places Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives in this topic. The required scope is: Configuration and control of DC motor drives, induction motor drives, permanent magnet motor drives, and switched reluctance motor drives.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system

named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Power Electronics, Drive Trains and Braking

9. Explain Power electronics converters used in electric and hybrid vehicles. [3 marks]

<p>Scope. The official syllabus places Power electronics converters used in electric and hybrid vehicles in this topic. The required scope is: Rectifiers, inverters, DC/DC converters, and power-split devices for hybrid vehicles.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Power electronics converters used in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Control systems for EV. **[3 marks]**

<p>Scope. The official syllabus places Control systems for EV in this topic. The required scope is: Sizing the drive system, propulsion motor, power electronics, energy-storage technology, communications, and supporting subsystems.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Control systems for EV</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Drive trains in Electric and Hybrid Electric Vehicles. **[3 marks]**

<p>Scope. The official syllabus places Drive trains in Electric and Hybrid Electric Vehicles in this topic. The required scope is: Basic architecture, general EV configuration, alternatives based on drive-train and power-source configuration, single/multi-motor and in-wheel drives.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Drive trains in Electric and Hybrid Electric Vehicles</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Drive train configurations in electric and hybrid electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Drive train configurations in electric and hybrid electric vehicles in this topic. The required scope is: Series, parallel, series-parallel, and complex hybrid drive trains.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Energy Storage, Fuel Cells and Charging

13. Explain Energy storage requirements in electric and hybrid vehicles. [3 marks]

<p>Scope. The official syllabus places Energy storage requirements in electric and hybrid vehicles in this topic. The required scope is: Energy-storage requirements and battery parameters for EV and HEV applications.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Energy storage requirements in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical

implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Construction and working of batteries used in electric and hybrid vehicles. **[3 marks]**

<p>Scope. The official syllabus places Construction and working of batteries used in electric and hybrid vehicles in this topic. The required scope is: Lead-acid, nickel-based, sodium-based, lithium-based, and metal-air batteries; construction, operation, and characteristics.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Construction and working of batteries used in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages. **[3 marks]**

<p>Scope. The official syllabus places Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages in this topic. The required scope is: Features and principle of ultracapacitors, flywheel storage, and hybridization of storage devices.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Operation of Fuel cells. **[3 marks]**

<p>Scope. The official syllabus places Operation of Fuel cells in this topic. The required scope is: Basic principle, operation, and types of fuel cells.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Operation of Fuel cells</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Need for hybrid and electric vehicles* with one relevant application. (5 marks)
2. **2.** Define or explain *Components and working principle of electric and hybrid electric vehicles* with one relevant application. (5 marks)
3. **3.** Define or explain *Design requirements for electric vehicles* with one relevant application. (5 marks)
4. **4.** Define or explain *Performance during acceleration, coasting, moving up and down a hill* with one relevant application. (5 marks)
5. **5.** Define or explain *Power electronics converters used in electric and hybrid vehicles* with one relevant application. (5 marks)
6. **6.** Define or explain *Control systems for EV* with one relevant application. (5 marks)
7. **7.** Define or explain *Energy storage requirements in electric and hybrid vehicles* with one relevant application. (5 marks)
8. **8.** Define or explain *Construction and working of batteries used in electric and hybrid vehicles* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Electric and Hybrid Vehicle Fundamentals:** Electric vehicle architecture and policy comparison.
- **EV Design and Motor Drives:** Use tractive effort and motor characteristics to justify a drive choice.
- **Power Electronics, Drive Trains and Braking:** Trace power flow from battery through converter and motor to wheels, including regenerative return.
- **Energy Storage, Fuel Cells and Charging:** Select storage and charging technology against range, power, safety, and infrastructure constraints.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Need for hybrid and electric vehicles	<p>Scope. The official syllabus places Need for hybrid and electric vehicles in this topic. The required scope is:

Term	Short meaning
	Need for hybrid and electric vehicles; historical background; social and environmental importance. Conce
Components and working principle of electric and hybrid electric vehicles	<p>Scope. The official syllabus places Components and working principle of electric and hybrid electric vehicles in this topic. The required scope is: Main components and working principles of electric and hybrid vehicles.</p> <p><str
Configurations of electric vehicles and hybrid vehicles	<p>Scope. The official syllabus places Configurations of electric vehicles and hybrid vehicles in this topic. The required scope is: Pure Electric vehicle (PE), Battery Electric vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid
Compare petrol, diesel, hybrid and electric vehicles	<p>Scope. The official syllabus places Compare petrol, diesel, hybrid and electric vehicles in this topic. The required scope is: Comparative study of EV and conventional vehicles; differences between complete EV and hybrid vehicles;
Advantages and disadvantages of hybrid and electric vehicles	<p>Scope. The official syllabus places Advantages and disadvantages of hybrid and electric vehicles in this topic. The required scope is: Technical, economic, environmental, range, charging, maintenance, and lifecycle advantages and l
Government policies and schemes related to Electric vehicles	<p>Scope. The official syllabus places Government policies and schemes related to Electric vehicles in this topic. The required scope is: FAME-1, FAME-2, Transformative Mobility and Energy Storage Mission, central and state policies,
Design requirements for electric vehicles	<p>Scope. The official syllabus places Design requirements for electric vehicles in this topic. The required scope is: Range, maximum velocity, acceleration, power requirement, mass, resistances, tractive effort, transmission requirem
Performance during acceleration, coasting, moving up and down a hill	<p>Scope. The official syllabus places Performance during acceleration, coasting, moving up and down a hill in this topic. The required scope is: Performance of electric vehicles during acceleration, coasting, and motion up or down a
Types of motors used in electric vehicles	<p>Scope. The official syllabus places Types of motors used in electric vehicles in this topic. The required scope is: Classification of EV motors and selection considerations.</p> <p> Concept and method. Study this to
Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives	<p>Scope. The official syllabus places Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives in this topic. The required scope is: Configuration and control of DC motor drives, induction
Power electronics converters used in electric and hybrid vehicles	Scope. The official syllabus places Power electronics converters used in electric and hybrid vehicles in this topic. The required scope is: Rectifiers, inverters, DC/DC converters, and power-split devices for hybrid vehicles. <

Term	Short meaning
Control systems for EV	<p>Scope. The official syllabus places Control systems for EV in this topic. The required scope is: Sizing the drive system, propulsion motor, power electronics, energy-storage technology, communications, and supporting subsystems.</p
Drive trains in Electric and Hybrid Electric Vehicles	<p>Scope. The official syllabus places Drive trains in Electric and Hybrid Electric Vehicles in this topic. The required scope is: Basic architecture, general EV configuration, alternatives based on drive-train and power-source config
Drive train configurations in electric and hybrid electric vehicles	<p>Scope. The official syllabus places Drive train configurations in electric and hybrid electric vehicles in this topic. The required scope is: Series, parallel, series-parallel, and complex hybrid drive trains.</p> <p>Concep
Regenerative braking and hybrid braking	<p>Scope. The official syllabus places Regenerative braking and hybrid braking in this topic. The required scope is: Braking in electric and hybrid vehicles, energy recovery and coordination with friction braking.</p> <p>Conce
Service and maintenance of EV components	<p>Scope. The official syllabus places Service and maintenance of EV components in this topic. The required scope is: Maintenance, repair, service, and troubleshooting of motor, drive train, battery, and other EV components.</p> <p><s
Energy storage requirements in electric and hybrid vehicles	<p>Scope. The official syllabus places Energy storage requirements in electric and hybrid vehicles in this topic. The required scope is: Energy-storage requirements and battery parameters for EV and HEV applications.</p> <p>Co
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Operation of Fuel cells	<p>Scope. The official syllabus places Operation of Fuel cells in this topic. The required scope is: Basic principle, operation, and types of fuel cells.</p> <p> Concept and method. Study this topic as a sequence of <s
Battery charging Technologies and Battery management systems	<p>Scope. The official syllabus places Battery charging Technologies and Battery management systems in this topic. The required scope is: Onboard/offboard charging, domestic/public infrastructure, fast charging, battery swapping, and

Official References and Resources

References listed in the supplied syllabus

1. Jack Erjavec and Jeff Arias, Hybrid, Electric and Fuel-cell Vehicles, Cengage Learning India Pvt Ltd.
2. C. Mi, M. A. Masrur and D. W. Gao, Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives, Wiley, 2011.
3. S. Onori, L. Serrao and G. Rizzoni, Hybrid Electric Vehicles: Energy Management Strategies, Springer, 2015.
4. M. Ehsani et al., Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, CRC Press, 2004.
5. T. Denton, Electric and Hybrid Vehicles, Routledge.
6. Iqbal Hussein, Electric and Hybrid Vehicles Design Fundamentals, CRC Press.
7. A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, 2018.

Official learning resources listed

- https://nptel.ac.in/courses/108/103/108103009/
- https://nptel.ac.in/courses/108/106/108106170

In-page Search